

EN4730 Convex Engineering Design Assignment 02

M. T. U. Sampath K. Perera

March 28, 2026

Objective

In this assignment, you will explore the connection between saddle points of the Lagrangian and the concept of strong duality, using a game-theoretic perspective.

1. Follow the steps below to analyze and interpret the mini-max inequality and its implications in the context of optimization and zero-sum games.

Understand the game setup.

- Suppose two teams are involved in a cricket match with payoff function $f(w, z)$, where $f(w, z)$ represents the total runs scored by Team B in an over.
- Team A (bowling) chooses $w \in W$ to *minimize* runs.
- Team B (batting) chooses $z \in Z$ to *maximize* runs.
- For simplicity, we approximate this over-level interaction as a zero-sum game: Team B's gain in runs corresponds directly to Team A's loss.

(a) What is the meaning of $\inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right)$? [05 marks]

(b) What is the meaning of $\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right)$? [05 marks]

Analyze the order of play.

- Consider two scenarios: (a) Team A moves first, and (b) Team B moves first.
(a) Use reasoning to explain why the inequality

$$\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right) \leq \inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right)$$

always holds.

[20 marks]

- (b) Consider a single over. Suppose Team A has two bowling strategies: Defensive(w_1) and Aggressive(w_2), and Team B has two batting strategies: Conservative(z_1) and Aggressive(z_2). The payoff table (runs scored by Team B in an over) is:

	z_1 (Conservative)	z_2 (Aggressive)
w_1 (Defensive)	4	10
w_2 (Aggressive)	2	12

i. **Minimax:**

If Team A bowls first, compute $\inf_w \left(\sup_z f(w, z) \right)$ and explain its meaning in the context of the over. **[15 marks]**

ii. **Maximin:**

If Team B bats first, compute $\sup_z \left(\inf_w f(w, z) \right)$ and explain its meaning in the context of the over. **[15 marks]**

iii. **Saddle Point:**

Verify the minimax inequality $\sup_z \left(\inf_w f(w, z) \right) \leq \inf_w \left(\sup_z f(w, z) \right)$ and determine if a saddle point exists. **[25 marks]**

2. Write-down the equivalent linear programming (LP) of the following optimization problems.

(a)

$$\begin{aligned} & \text{minimize } \|\mathbf{x}\|_1, \\ & \text{subject to } \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_\infty \leq 1. \end{aligned}$$

(b)

$$\text{minimize } \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_1.$$

[15 marks]

Submission

- Upload a report as a pdf file named as "yourindexno_EN4730_A02.pdf". Include the index number and the name within the report as well.
- The interpretation of results and the discussion are important in the report.
- Pay careful attention to formatting such as font size, spacing, and margins.
- Include a title page with necessary information (e.g., title, author, date, index no).
- Use consistent and professional formatting throughout the document.
- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism¹ or cheating will not be tolerated.
- An extra penalty of 10% is applied for late submission.

¹<https://en.wikipedia.org/wiki/Plagiarism>